Pally: Groupware Heuristics Evaluation

Session A is the baseline (Discord voice only); Session B uses Discord voice with our group bot.

Coding key

Present	The mechanism exists and worked during testing.
Constrained	Supported, but limited; users met difficulty or needed a workaround.
Absent	The task needed the mechanism, but the system did not provide it.
N/A	The test tasks did not trigger this situation, so it cannot be judged.

Evaluation

#	Heuristic	Session A Discord voice only	Session B With Pally	Evidence
1	**Provide the means for intentional and appropriate verbal communication**	Present	Present	Session A Discord voice let participants talk to each other in real time. It was the only communication channel available. A and B talked frequently, while C talked only occasionally. Session B Discord voice and text channels let participants talk to each other in real time and discuss the outputs. Their talk frequency remained unchanged.
2	**Provide the means for intentional and appropriate gesture communication**	N/A	N/A	Session A The session had no gesture-based mechanism, and the tasks did not involve pointing at or indicating a person or object. Session B Pally has no gesture-based mechanism either.
3	**Provide consequential communication of an individual's embodiment**	Present	Present	Session A Discord's voice channel shows who has joined, who is speaking and who is muted, and voice tone adds further cues. Session B This was provided mainly by Discord: online status, voice channel membership and mute state show who is present. Pally adds no further embodiment information.

#	Heuristic	Session A Discord voice only	Session B With Pally	Evidence
4	**Provide consequential communication of shared artifacts**	N/A	Constrained	Session A No shared artifact was used in this session. Session B Pally's outputs became shared information. For example, once Random Pick selected C, A and B immediately recognized the result. However, by design, most of Pally's outputs are anonymous. Participants can hardly tell who triggered a command or who submitted an evaluation; they see only the result.
5	**Provide protection**	Absent	Present	Session A Without a tool, everything was said openly by voice. When C had to give opinions in front of A and B, C had no way to stay anonymous or soften the exposure. Session B Pally provided an anonymous way to express opinions. Other participants cannot identify who submitted an evaluation or triggered a tool.
6	**Management of tightly and loosely-coupled collaboration**	Constrained	Present	Session A Voice supports only real-time, synchronous talk (tight coupling). There was no way for A, B, and C to contribute asynchronously or to work separately and merge later. Session B Voice and Random Pick support immediate, synchronous use, while Trait Picker supports asynchronous submission followed by a shared summary. Participants moved between the two modes smoothly.
7	**Allow people to coordinate their actions**	Constrained	Present	Session A Coordination happened through purely verbal negotiation, for example, when A, B, and C decided who would be the decision maker. There was no shared mechanism, so it relied entirely on talking. Session B Besides voice negotiation, Pally gave the group a shared, neutral way to decide who would be selected without negotiating among themselves, and it reallocated the decision-making power.
8	**Facilitate finding collaborators and establishing contacts**	N/A	Absent	Session A Participants were already together in the same voice channel, so finding or inviting collaborators was not part of the session. Session B By design, all participants remain anonymous when using Pally's three tools, so Pally does not support finding collaborators.